

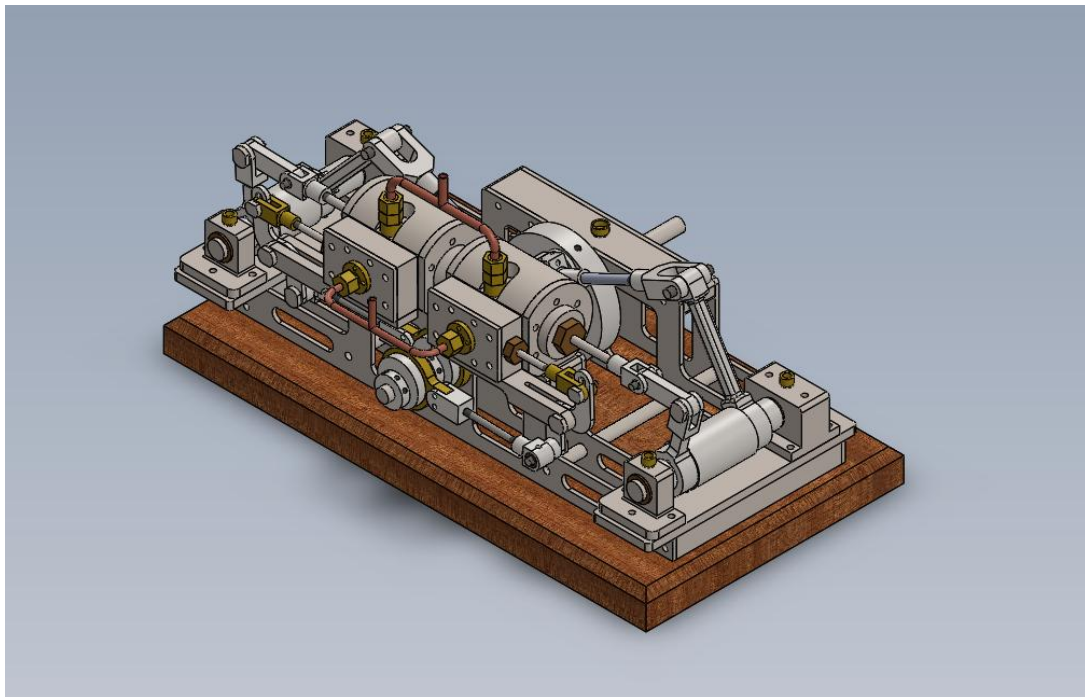
Horizontal Steam Engine – Monitor Type

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1. Title

Design and Assembly of a Horizontal Steam Engine (Monitor Type) Using SOLIDWORKS



2. Objective

The objective of this project was to design, model, and assemble a horizontal steam engine of the Monitor type using SOLIDWORKS. This included creating each component as an individual 3D model, assembling the engine with proper constraints, and preparing a complete engineering drawing that clearly communicates the structure and function of the system.

3. Description of the Project

3.1 Overview

This project recreates a Horizontal Steam Engine of the Monitor Type, a classic single-cylinder design widely used in early industrial machinery. The engine is known for its simple

mechanical layout: a horizontally positioned cylinder, a piston-and-rod mechanism, and a crankshaft that converts reciprocating motion into rotation. The design demonstrates core engineering concepts such as slider–crank motion, alignment, and load-bearing structure.

The final assembly contains more than seventy individual components, including the cylinder assembly, piston rod, crankshaft, bearings, guides, frame structure, connecting links, and various fasteners. All components were assigned practical engineering materials such as aluminium, cast iron, copper, brass, bronze, and stainless steel, reflecting realistic mechanical properties, appearance, and manufacturing considerations.

The mechanical operation is fully represented in the assembly: the piston travels within the cylinder, the connecting rod transfers this motion to the crankshaft, and the crankshaft produces rotational output. Structural elements like supports, guides, and mounting plates ensure stability and proper alignment. The overall design reflects both the functional behaviour and the characteristic appearance of traditional Monitor-type steam engines.

3.2 Part Modelling

Each part was created individually in SOLIDWORKS using tools such as extrudes, revolves, fillets, chamfers, cuts, and patterns. Components were dimensioned to ensure accurate fit and realistic proportions. Material assignments—aluminium, brass, bronze, cast iron, stainless steel, and copper—were applied to represent real-world construction.

3.3 Assembly Modelling

After completing all parts, they were assembled using concentric, coincident, and distance mates to correctly position the crankshaft, piston, connecting rod, and structural members. The assembly accurately represents the motion and mechanical interactions of the engine.

3.4 Engineering Drawings

A comprehensive engineering drawing was prepared from the final assembly. It includes orthographic views, a detailed Bill of Materials, item balloons, and a standard title block, providing clear documentation of the engine model.

4. Conclusion

The project successfully models a Horizontal Steam Engine of the Monitor Type using SOLIDWORKS. With over seventy parts designed, assigned realistic materials, and assembled into a complete working model, the project demonstrates strong understanding of 3D part design, mechanical assembly, and technical drawing practices. The references used for this project were taken from drawing blueprints by JDW Draughting Services, New Zealand.